AI Trip Planner - Comprehensive Project Report

Project Title: AI-Powered Personalized Trip Planner

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1. Introduction

The AI Trip Planner is a state-of-the-art web application designed to simplify travel planning. By leveraging Artificial Intelligence, the platform generates tailored itineraries based on user preferences such as destination, budget, travel style, and specific interests. The application focuses on a premium user experience with cinematic visuals and realistic, actionable travel data.

2. Core Features

- AI Itinerary Generation: Real-time generation of day-wise plans using Google Gemini.

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3. Technology Stack

Frontend

- React.js: Component-based UI development.

- Vite

Backend (AI Services)

- FastAPI (Python): Modern, fast web framework for building APIs.

- Goo

Backend (User Authentication)

- PHP: Server-side scripting for database interactions.

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4. System Architecture & Workflow

Workflow Steps:

1. User Interaction: User submits a trip request via the React frontend.

2. AI
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5. Detailed File Explanations

Frontend Components

- App.jsx: Handles global state and client-side routing.

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Backend Services

- ai_service.py: Contains the logic for prompt engineering and AI communication.

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6. Database Design (MySQL)

The system uses a relational database to manage users.

Table: users

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7. Core Logic Implementation

AI Prompt Engineering (`ai_service.py`)

The system uses a specific instruction set to ensure the AI returns data in a usable format:

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Authentication Logic (`register.php`)

Ensures security by hashing passwords before storage:

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8. Installation & Setup

1. Frontend:

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9. Conclusion

The AI Trip Planner demonstrates the powerful integration of Generative AI with modern web technologies. It provides a scalable solution for personalized travel planning, offering a glimpse into the future of automated concierge services.